

Guide: The Verified Residue-Coverage Tool

Collatz proof project

September 5, 2026

What has been proved

Every positive Collatz orbit visits a number whose remainder on division by nine is two. Such visits occur after any specified time. This is a formalization of a known theorem of Monks and coauthors, not a new proof of the Collatz conjecture.

How the argument works

First move past any multiples of three. The remaining remainders are one, two, four, five, seven, and eight. If the orbit avoids two, the allowed remainder transitions force it eventually to stay at eight. An even number with remainder eight moves to remainder four, so this would require only odd steps forever. Our existing Lean parity argument rules that out for natural starting values.

What consecutive visits tell us

Between consecutive visits to the progression, at most six steps are even. The number of odd steps can still be arbitrarily large. If the accumulated coefficient is below one, however, the return takes at most sixteen steps. This bound is attained: 147440 next visits at 132860 after sixteen steps. Lean also proves that every such contracting return from a seed above two has an endpoint smaller than its starting value. The proof combines a global arithmetic bound with a kernel-checked finite table, so it applies to every seed, not just those tested numerically. Growing returns exist too: eleven next visits at twenty. We have not proved that every orbit eventually takes a contracting return. The new theorem identifies the possible contracting branches of this return map; it does not establish convergence of that map.

Why the first-return condition matters

At a later visit, 470 reaches 479 with a coefficient below one. Lean checks this counterexample to the unrestricted extension. It had already descended earlier. A separate finite scan tested the first cumulative contraction observed at a visit, across 999,999 starting seeds. All tested endpoints descended, with no seed hitting the 3000-step cap. The longest case took 225 steps. That single case is also checked in Lean; the full scan remains experimental evidence. Neither universal descent at this sampled stopping time nor its existence for all seeds is proved.

The missing connection

Earlier work guarantees noncontracting starting points on hypothetical unbounded orbits. Residue coverage does not say those points are two modulo nine. These two kinds of visits have not been shown to coincide. The Collatz conjecture remains open.

Verification

From `lean/`, run `lake build Collatz.ResidueCoverage`. From the repository root, run `python3 analysis/audit_theorems.py --build` with the project's Lean executable on `PATH`. The selected audit checks the actual theorem statements and axiom dependencies; no additional axiom asserts the literature result.